

Multiplicative Obstructions at Base 3/2

Lean-checked exclusions for the linear-form approach to Erdős Problem #1049

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ABSTRACT

Let $F(t) = \sum_{n \geq 1} (t^n - 1)^{-1}$. At the rational base $3/2$ we prove exclusions for the approach to irrationality by integer linear forms, and formalise them in Lean. Rowwise content (the greatest common divisor of the two entries of a coefficient row) scales the analytic error and the exterior determinant by exactly the same factors, including the absolute determinant height, so a local divisor supplied only by the two row contents is paid for by exactly the same factor in the Archimedean determinant height. Homogeneous specialisation at $(3, 2)$ sees the constant endpoint modulo 3 and the top endpoint modulo 2, so unit endpoints exclude both primes from any common divisor. Together: when one entry of a coefficient pair has a unit top endpoint and the other a unit constant endpoint, neither rowwise scalar content nor a common divisor of the two specialised evaluations can manufacture the two local primes required at $3/2$. The missing local gain must therefore survive primitive normalisation. What remains is a growing-rank additive kernel cancelling both endpoint jets (the residues of a specialised coefficient modulo 3^R and modulo 2^S) in both specialised coefficient sequences while preserving a nonzero, sufficiently small remainder; we count the four-jet target exactly and prove the concrete sufficient threshold $M \geq 4R + 2S$ on the number M of coefficient pairs, at positive bottom depth R .

We also record what the elementary route costs. Erdős's argument for the irrationality of $\sum_{n \geq 1} c(n)b^{-n}$ at an integer base $b \geq 2$ proceeds by clearing denominators at a cut level and trapping a positive integer below 1. At a rational base r/s with $s \geq 2$ the same clearing leaves a factor s^{N+1} behind. Writing a coordinatewise corridor for the surviving arithmetic, we prove that a corridor forces $s^{N+K+1} < r(N+K)$, and that at the base $3/2$ this is impossible once the shift and the cleared window are both nonempty. We also prove the exact denominator-cleared tail recurrence $U_{N+1} = rU_N - Bc(N+1)s^{N+1}$ and the bound $2^{N+1} \leq Bc(N+1)s^{N+1}$ for $s \geq 2$ and positive data, against the exact collapse of that term to $Bc(N+1)$ at $s = 1$. The height criterion of Bundschuh and Väänänen (1994), which settles a restricted family of rational bases containing $7/2$, is used as an external theorem. We formalise the complete elementary Archimedean height inequality required at $7/2$, while leaving the source theorem's analytic irrationality argument external. Separately we check polynomial bounds on the denominator exponents of a homogenised Padé construction, with the analytic remainder left untouched.

Problem #1049 is open, and nothing here decides it. The results are exclusions of particular constructions, not statements about the arithmetic nature of any value: we do not prove irrationality at $3/2$, at any other rational base, or in the unrestricted problem.

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1 Introduction

Let $t > 1$ be a rational number and let $\tau(n)$ count the divisors of n . Erdős Problem #1049 asks whether

$$F(t) = \sum_{n \geq 1} \frac{1}{t^n - 1} = \sum_{n \geq 1} \frac{\tau(n)}{t^n}$$

is irrational [2, p. 102]. The two forms agree by expanding $(t^n - 1)^{-1} = \sum_{k \geq 1} t^{-nk}$ and collecting the terms with the same exponent, the coefficient of t^{-m} being the number of divisors of m . The question is a conjecture of Chowla; Erdős proved it for every integer $t \geq 2$ [1]. Numbering and status follow Bloom's catalogue [10]. For non-integer rational t the problem is open.

Write $t = r/s$ in lowest terms with $r > s \geq 1$, so that $s = 1$ is exactly the integer case Erdős settled. The resistant explicit base of least naive height $H(r/s) = \max(r, s)$ is $t = 3/2$. A published height criterion of Bundschuh and Väänänen [4, Thm. 2, pp. 176–177] settles a family of rational bases restricted by a height condition; that family contains $7/2$ and does not contain $3/2$. Between the two lies the question this note is about: what exactly stops the integer-base argument from running at $3/2$?

Relation to prior work. The *Formal Conjectures* file for Problem #1049 contains the conjecture and integer-base theorem as sorry placeholders, but it also proves the Lambert identity between the two displayed series for rational t [11]. In the present $t > 1$ regime its `lambert_convergent` branch is an ordinary convergent-series proof; the same file's `|t| ≤ 1` branch instead uses Lean's convention that the `tsum` of a nonsummable series is zero. Thus it supplies genuine checked prior art for the identity, but no irrationality theorem. The present note formalises propositions about the clearing argument and likewise does not answer the conjecture. Erdős's positive-integer theorem of 1948 [1] sits inside a larger integer-base literature. In 1991 Borwein proved the irrationality of shifted series $\sum_{n \geq 1} (t^n + w)^{-1}$ at integer bases $t \geq 2$, for every nonzero rational w with $w \neq -t^m$ for all $m \geq 1$, by Padé approximation rather than by digit clearing; his estimates also show that these values are not Liouville numbers [3, Thm. 4, pp. 257–258]. In 2001 Van Assche recovered the integer-base irrationality and an irrationality-measure bound using little q -Legendre Padé approximants [7, Thm. 1, p. 10; proof pp. 10–11]. His more general Theorem 3 proves irrationality of $\sum_{k \geq 1} (cp^k - 1)^{-1}$ for an integer $p > 1$ and fixed rational c away from the poles [7, Thm. 3, p. 14]; it does not cover a rational noninteger base t in $F(t)$, because the multiplier needed to write t^k over an integer base varies with k . In 2013 Vandehey proved that $\sum_{n \geq 1} d(n)a_n/b^n$ is irrational whenever $b > 1$ is an integer and (a_n) ranges in a finite integer alphabet excluding zero; taking $a_n = (-1)^n$ completes the elementary digit method for negative integer bases [8, Thm. 1.2, p. 2]. All three retain an integer base. In 2004 Zudilin obtained integer-base irrationality-exponent estimates for q -harmonic series on a parameter cone, by way of Heine's basic transform and a permutation group [5, Secs. 4–5, pp. 159–162]. The ordinary-hypergeometric antecedent is Rhin and Viola's S_5 action and twelve-coset denominator reduction for rational forms in $\zeta(2)$ [6, Sec. 3, pp. 38–42; Sec. 4, pp. 46–51]. The coefficient polynomials specialised in Section 3 are those of that construction. The neighbouring problem of Lambert sub-series $\sum_{n \in A} (t^n - 1)^{-1}$ over a restricted index set A is treated by Kovač and Tao [9]. The

rational non-integer progress relevant here is instead the height criterion of Bundschuh and Väänänen [4]. No claim of priority is made for anything below, which concerns the formal status of an argument rather than a new theorem about $F(t)$.

The integer-base argument, in outline. Let $b \geq 2$ be an integer and suppose $F(b) = p/q$ for integers p and $q \geq 1$. Cut the series at a level N , multiply by qb^N , and subtract the cleared prefix. What remains is a positive integer, since every term of the prefix has been cleared and the tail is positive; and it is smaller than 1, since the tail of a geometric-type series is small. A positive integer below 1 does not exist. The argument is elementary and it is sensitive in exactly one place: the clearing must remove every denominator.

Write $\beta = r/s$ for the base and $c(n)$ for the coefficient of β^{-n} , so that $c = \tau$ in the case at hand. The term $c(n)\beta^{-n}$ is $c(n)s^n/r^n$. Clearing the power of r leaves the numerator factor s^n in place. That factor is invisible when $s = 1$ and grows geometrically when $s \geq 2$. At 3/2 it is 2^n .

Terminology. The linear-form route works with pairs (U, V) , of integers or of integer polynomials; following matrix usage we call such a pair a *row*, and its *content* is the greatest common divisor of its two entries. A row of content 1 is *primitive*, and dividing a row by its content is *primitive normalisation*. The *exterior determinant* of two integer rows is $U_n V_m - U_m V_n$, the determinant of the 2×2 matrix they form. Two quantities attached to that determinant are compared throughout: an integer dividing it, which is a local gain, and its absolute value, which is an Archimedean cost; we call that comparison the *local-to-Archimedean balance*. The *endpoints* of a coefficient polynomial, taken relative to the declared width W of Section 3, are its constant coefficient and its coefficient at W ; we call these the *constant endpoint* and the *top endpoint*, so the top endpoint is the coefficient at W and not the leading coefficient unless the two agree. A *unit endpoint* is one equal to ± 1 , the units of $\mathbb{Z}$; Theorem 3.1 is the reason only these two coefficients decide divisibility by 3 and by 2 after specialisation at $(3, 2)$. A *jet* is a residue of a specialised coefficient modulo a prime power: the bottom jet is its residue modulo 3^R and the top jet its residue modulo 2^S , and R and S are the bottom and top *depths*. A jet vanishes exactly when the prime power in question divides the specialised coefficient.

The shortfall at 3/2. The elementary route and the linear-form route are both examined below. For the coordinatewise clearing scheme the leftover at each step is the forcing term of an exact recurrence, of size at least 2^{N+1} whenever $s \geq 2$ and the scaling constant B and the coefficient $c(N+1)$ are at least 1 (Theorem 5.2), and the scheme itself is excluded at 3/2 for every shift $N \geq 1$ and every cleared window of width $K \geq 1$ (Theorem 4.5). For the linear-form constructions the shortfall is the 2-adic and 3-adic gain that has to remain after the coefficient rows have been divided by their contents (Section 2). Each multiplicative device examined here for supplying that gain is excluded: rowwise content is exactly neutral, since it scales the exterior determinant and its absolute height by the same factor (Theorem 2.1), and unit endpoints keep both 2 and 3 out of any common divisor of the two specialised evaluations (Theorem 3.1 and Proposition 3.3). Corollary 3.5 states the two together. Separately, the scalar parameter margin is negative under the assumed source inequality (Theorem 3.8). The device pursued here instead is additive: an integer relation among rows that cancels the endpoint

jets. Theorem 3.6 shows that a nonzero relation with coefficients in $\{-1, 0, 1\}$ cancelling all four jets exists whenever the bottom depth is positive and the number of coefficient pairs is at least $4R + 2S$. It does not show that the resulting combination has a nonzero polynomial pair or a nonzero remainder, which is what Problem 8.1 asks for. Sections 6 and 7 record two external routes and what each leaves unproved at 3/2.

Sharpness. Two questions of scope are worth isolating. The corridor bound of Theorem 4.2 is exponential on the left and linear on the right, so for a fixed numerator and any $s \geq 2$ a corridor can survive only for bounded $N + K$; the base 3/2 is the case in which the crossing has already happened at the smallest admissible window, which is why the exclusion there holds for all $N \geq 1$ and $K \geq 1$ with no further restriction. The height criterion of [4] is restricted by a height condition satisfied at 7/2 and not at 3/2, so the two bases are separated by that criterion rather than by anything proved here.

Status. The problem treated here is open, and this note does not close it. Every statement below marked as checked is a proposition that the pinned Lean kernel accepts from the sources this note links to, with no sorry, no added axiom, and no unchecked evaluation. That is a claim about the formal statement, not about its mathematical interest, its novelty, or the original problem. The unresolved obligations are named exactly, in their own section, and none of the finite computations, reductions, or no-go results here removes one of them.

Structure. Section 2 proves that rowwise content is neutral for the local-to-Archimedean balance. Section 3 proves the endpoint congruences at (3,2), deduces from them and from Section 2 that neither rowwise content nor a common divisor of the two specialised evaluations supplies the missing local gain, gives the four-jet collision count, and records one further exclusion on Zudilin’s scalar parameters. Sections 4 and 5 return to the elementary clearing scheme and record what the residue s^n costs there, first as an exclusion and then as an exact recurrence with a lower bound on the surviving term. Sections 6 and 7 record what is and is not formalised of two external routes. Section 8 states the remaining obligations. Linked phrases open the corresponding Lean declaration at the pinned source revision 08d83b6689c8.

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2 Row content does not change the local-to-Archimedean balance

An irrationality argument by linear forms replaces the clearing scheme by explicit rational approximation: one constructs integer linear forms in 1 and $F(\beta)$ whose analytic decay outruns the height of their common denominator. For an integer coefficient pair (U, V) and a real target S , put

$$L_S(U, V) = US - V, \quad \Delta((U_n, V_n), (U_m, V_m)) = U_n V_m - U_m V_n.$$

We call $L_S(U, V)$ the *error* of the row at S ; a good approximation is one that makes it small. The second expression is the exterior determinant of the two rows, and it elimi-

Statement	Status	Treatment here
Irrationality of $F(3/2)$	Open	Not proved, and no partial result here bears on it.
Irrationality at some rational non-integer base	Proved elsewhere	The height criterion of [4]; cited, not formalised.
Row content is neutral for the local-to-Archimedean balance	Proved here	Theorem 2.1, including the absolute determinant height.
Homogeneous endpoint residues at $(3, 2)$	Proved here	Theorem 3.1.
Common divisor of the two specialised evaluations avoids 2 and 3	Proved here	Proposition 3.3; under unit endpoint hypotheses.
No gain from row content or from a common divisor at 3/2	Deduced here	Corollary 3.5; an ordinary deduction from the two preceding rows, not separately formalised.
Four-jet collision at $M \geq 4R + 2S$ coefficient pairs	Proved here	Theorem 3.6; sufficient, at positive bottom depth.
Scalar margin on Zudilin's parameter cone	Negative under source input	Theorem 3.8; the source inequality is assumed.
Corridor forces $s^{N+K+1} < r(N + K)$	Proved here	Theorem 4.2.
No corridor at base 3/2	Proved here	Theorem 4.5; a statement about the clearing scheme.
Cleared-tail recurrence	Proved here	Theorem 5.1, an exact identity.
Forcing term at least 2^{N+1} for $s \geq 2$	Proved here	Theorem 5.2; at $s = 1$ the term is $Bc(N + 1)$.
The elementary 7/2 height condition	Proved here	Theorem 6.1; the analytic theorem of [4] remains external.
Padé denominator-exponent bounds	Proved here	Section 7; exponent arithmetic only.
Padé remainder positivity and decay	Not treated	Section 7.
Growing-rank simultaneous endpoint-jet kernel	Open	Section 8; the exact remaining obligation.

nates S exactly:

$$\Delta = U_m L_S(U_n, V_n) - U_n L_S(U_m, V_m).$$

This identity is what makes the determinant useful. Since Δ is an integer, if it does not vanish then

$$1 \leq |\Delta| \leq |U_m| |L_S(U_n, V_n)| + |U_n| |L_S(U_m, V_m)|,$$

and the two errors cannot both be smaller than $1/(|U_n| + |U_m|)$. The determinant therefore carries two competing quantities at once, an integer that divides it and its own absolute value, and the theorem below is about how a rescaling moves them.

Theorem 2.1 (row-content no-go). *Let S be real, let (U_n, V_n) and (U_m, V_m) be pairs of integers, and let c_n, c_m be integers. Then*

$$L_S(c_n U_n, c_n V_n) = c_n L_S(U_n, V_n),$$

$$\Delta(c_n(U_n, V_n), c_m(U_m, V_m)) = c_n c_m \Delta((U_n, V_n), (U_m, V_m)),$$

and consequently

$$|\Delta(c_n(U_n, V_n), c_m(U_m, V_m))| = |c_n| |c_m| |\Delta((U_n, V_n), (U_m, V_m))|.$$

In particular $c_n c_m$ divides the scaled determinant. Hence a local divisor supplied only by the two row contents is paid for by exactly the same factor in the Archimedean determinant height.

Proof. All three identities are routine expansions in $\mathbb{Z}$ or $\mathbb{R}$; the divisibility statement uses the primitive determinant as its witness. $\square$

Informally, Theorem 2.1 says that a common factor introduced into the rows appears twice over: once as a divisor of the exterior determinant, and once as the same factor in the absolute value of that determinant. It moves the local gain and the Archimedean cost by exactly the same amount.

Example 2.2. Take $(U_n, V_n) = (1, 2)$ and $(U_m, V_m) = (3, 5)$, so that $\Delta = 1 \cdot 5 - 3 \cdot 2 = -1$. Multiplying the first row by $c_n = 6$ and the second by $c_m = 10$ gives the rows $(6, 12)$ and $(30, 50)$, whose determinant is $6 \cdot 50 - 30 \cdot 12 = -60$. That determinant is now divisible by 60, which looks like a local gain of 60; and its absolute value has risen from 1 to 60, which is a cost of exactly the same size.

Lean checks the error identity in [error scaling](#), the determinant identity in [content factorisation](#), the exact absolute-height identity in [absolute determinant scaling](#), and the divisor statement in [content-product divisibility](#). The elimination identity is the checked [exterior determinant identity](#).

The key point is that the two scalings are the same scaling. Each identity on its own is a one-line expansion, and the interest of the theorem is not in any one of them. Taken together they say that the factor $c_n c_m$ which a rescaling introduces into the determinant reappears undiminished, as $|c_n| |c_m|$, in the absolute value of that determinant. The third identity is displayed with absolute values for exactly that reason: it is what makes the statement one about the Archimedean height and not about divisibility alone. Whatever c_n and c_m are, a rescaling therefore leaves the balance between the local divisor and that height where it was.

The theorem does not construct primitive Padé rows, estimate their remainders, or prove that their exterior determinant is nonzero. It removes one source of apparent gain: multiplying a useful row by a large common integer cannot improve the local-to-Archimedean balance. Every candidate family must first be divided by its rowwise common contents; the required 2-adic and 3-adic gain must remain afterwards. The theorem quantifies over arbitrary integer coefficient pairs, so it applies to the rows of any construction of this shape, whatever produced them.

3 Endpoint residues at (3, 2) and the four-jet kernel

Zudilin's treatment of q -harmonic series [5] builds linear forms of the shape used in Section 2 out of Heine's basic transform. What matters here is one feature of that construction: its coefficients are produced as polynomials with integer coefficients, and a numerical base is substituted only at the end. We call these the Heine–Zudilin forms and ask what the substitution at 3/2 does to the two primes 2 and 3. The endpoint and jet arguments below use no property of the construction beyond integrality of those coefficient polynomials, and each of them quantifies over arbitrary elements of $\mathbb{Z}[X]$. The exception is Theorem 3.8 at the end of the section, which concerns the scalar parameters of Zudilin's cone rather than the coefficient polynomials.

Substituting $X = 3/2$ into an integer polynomial produces a rational number, and multiplying by a power of 2 clears its denominator. The following evaluation records that cleared numerator, so that all the arithmetic below stays inside $\mathbb{Z}$. For $P(X) = \sum_i p_i X^i \in \mathbb{Z}[X]$ and a declared width $W \geq 0$, put

$$H_W(P) = \sum_{i=0}^W p_i 3^i 2^{W-i}.$$

This is the [homogeneous endpoint evaluation](#). It is homogeneous in the sense that X^i is replaced by $3^i 2^{W-i}$, so the numerator and the denominator of the base are carried symmetrically. When $W \geq \deg P$ it is exactly the cleared numerator, since

$$\sum_{i=0}^W p_i 3^i 2^{W-i} = 2^W \sum_{i=0}^W p_i \left(\frac{3}{2}\right)^i = 2^W P\left(\frac{3}{2}\right).$$

Theorem 3.1 (endpoint residues). *Let $P = \sum_i p_i X^i \in \mathbb{Z}[X]$ and let $W \geq 0$. Then*

$$H_W(P) \equiv p_0 2^W \pmod{3}, \quad H_W(P) \equiv p_W 3^W \pmod{2}.$$

Consequently a unit constant coefficient prevents divisibility by 3, and a unit coefficient at the declared width W prevents divisibility by 2.

Proof. Modulo 3, every summand with $i > 0$ vanishes; modulo 2, every summand with $i < W$ vanishes. The remaining powers are units in the corresponding residue fields. $\square$

Since 2^W is invertible modulo 3 and 3^W is invertible modulo 2, the two congruences say more than the stated consequence: divisibility of $H_W(P)$ by 3 is decided by p_0 alone, and divisibility by 2 by p_W alone. The rest of the coefficient vector is invisible to both primes. The coefficient p_W is the top coefficient of P exactly when $\deg P = W$, and is zero when $\deg P < W$; the identity $H_W(P) = 2^W P(3/2)$ likewise holds only for $\deg P \leq W$.

Example 3.2. Take $W = 2$. The three polynomials below differ only at an endpoint.

P	$H_2(P)$	$3 \mid H_2(P)$	$2 \mid H_2(P)$
$X^2 + 1$	$1 \cdot 4 + 0 \cdot 6 + 1 \cdot 9 = 13$	no	no
$X^2 + 3$	$3 \cdot 4 + 0 \cdot 6 + 1 \cdot 9 = 21$	yes	no
$2X^2 + 1$	$1 \cdot 4 + 0 \cdot 6 + 2 \cdot 9 = 22$	no	yes

The first has both endpoints equal to 1 and its evaluation, 13, is divisible by neither prime; as a check, $2^2((3/2)^2 + 1) = 13$. The second and third show that each hypothesis is used: spoiling the constant endpoint admits 3, and spoiling the top endpoint admits 2.

The congruences are the [bottom-endpoint identity](#) and [top-endpoint identity](#); the unit consequences are the [constant-endpoint obstruction](#) and [top-endpoint obstruction](#).

The proof is two lines, but the shape of the statement is not incidental. The difficulty lies in the fact that the specialisation sees a different endpoint at each of the two primes: modulo 3 only the constant coefficient survives, and modulo 2 only the top one does. The two exclusions are therefore conditions at opposite ends of the coefficient vector, and the statement below imposes one at each end, on the two entries of a single coefficient pair.

Proposition 3.3 (common divisor). *Let $U, V \in \mathbb{Z}[X]$ and let $W \geq 0$. If U has unit top endpoint, V has unit constant endpoint, and an integer c divides both $H_W(U)$ and $H_W(V)$, then*

$$2 \nmid c \quad \text{and} \quad 3 \nmid c.$$

This is the checked [common-divisor exclusion](#). The proposition does not say that the two evaluations are coprime; it says that whatever they share misses both of the primes that matter at 3/2.

Example 3.4. Take $W = 2$, $U = X^2 + 3$ and $V = 5X^2 + 1$. The top endpoint of U and the constant endpoint of V are both 1, and

$$H_2(U) = 3 \cdot 4 + 1 \cdot 9 = 21, \quad H_2(V) = 1 \cdot 4 + 5 \cdot 9 = 49.$$

Here $\gcd(21, 49) = 7$, so a common divisor does exist and is not small; it is simply coprime to 6.

Corollary 3.5 (no gain from row content or a common divisor at 3/2). *Under the endpoint hypotheses of Proposition 3.3, neither rowwise scalar content nor a common divisor of the two specialised evaluations $H_W(U)$ and $H_W(V)$ can manufacture the two local primes required at 3/2.*

Proof. By Theorem 2.1 rowwise content multiplies the analytic error and the exterior determinant, including the absolute determinant height, by exactly the factors it introduces, so a divisor obtained that way is paid for by the same factor in that height. By Proposition 3.3 an integer dividing both specialised evaluations is divisible by neither 2 nor 3. $\square$

Both ingredients are Lean-checked; the combination is an ordinary deduction and is not separately formalised. The corollary excludes two ways of producing the required gain. It does not show that a gain of that kind is necessary for a proof by linear forms at 3/2.

Corollary 3.5 excludes the two multiplicative devices above, and what the rest of this section pursues in their place is additive: rather than multiplying one row by a scalar, take an integer combination of several rows and ask that the combination be divisible

where the individual rows are not. The endpoint congruences are the first case of a divisibility condition that can be imposed to any depth, and it is that condition, read additively, which is counted below.

We first raise the two congruences to prime powers. Fix depths $R, S \geq 0$. For $P \in \mathbb{Z}[X]$ the *bottom jet* $J_{3,R}(P)$ is the residue of $H_W(P)$ modulo 3^R , and the *top jet* $J_{2,S}(P)$ is its residue modulo 2^S ; Theorem 3.1 computes them at $R = S = 1$. Their vanishing is exactly the requested divisibility:

$$J_{3,R}(P) = 0 \iff 3^R \mid H_W(P), \quad J_{2,S}(P) = 0 \iff 2^S \mid H_W(P).$$

These are the checked [bottom-jet divisibility criterion](#) and [top-jet divisibility criterion](#). The *four-jet signature* of a coefficient pair (U, V) is then the quadruple

$$(J_{3,R}(U), J_{3,R}(V), J_{2,S}(U), J_{2,S}(V)) \in (\mathbb{Z}/3^R\mathbb{Z})^2 \times (\mathbb{Z}/2^S\mathbb{Z})^2,$$

two residues for each of the two primes, one from each entry of the pair. By the displayed criteria it vanishes exactly when 3^R divides both specialised entries and 2^S divides both.

This turns the missing local divisor into an additive congruence-kernel problem: what is sought is no longer a common divisor of the two evaluations, but a vector of small integer coefficients on which four residues vanish at once. Since H_W is linear in the coefficients of P , the four-jet signature of a combination is the corresponding combination of signatures, which is what makes the following count possible.

Theorem 3.6 (binary four-jet collision). *Fix a width W and depths R, S , and let $(U_j, V_j)_{j < M}$ be any M pairs of integral polynomials. Call a subset of $\{0, \dots, M-1\}$, equivalently a vector of $\{0, 1\}^M$, a binary selector. If the 2^M binary selectors outnumber the finite four-jet target*

$$(\mathbb{Z}/3^R\mathbb{Z})^2 \times (\mathbb{Z}/2^S\mathbb{Z})^2,$$

then two distinct subsets have the same four-jet sum. Subtracting their indicator vectors gives a nonzero coefficient vector in $\{-1, 0, 1\}^M$ cancelling all four jets. The target has exact cardinality

$$(3^R)^2(2^S)^2.$$

In particular, if $R > 0$ and $4R + 2S \leq M$, such a collision exists.

Proof. Send each binary selector to the sum of the four-jet signatures it selects. The claimed cardinal inequality and the pigeonhole principle give two distinct selectors in the same fibre. The cardinality formula is the product of the four cyclic-modulus cardinalities. For $R > 0$,

$$(3^R)^2(2^S)^2 < (4^R)^2(2^S)^2 = 2^{4R+2S} \leq 2^M,$$

which proves the stated sufficient threshold. □

The target count is the checked [four-jet target cardinality](#); the abstract collision is the checked [four-jet pigeonhole kernel](#), and the linear sufficient condition is the checked [rank–depth collision threshold](#). The count is a routine pigeonhole; the reformulation

above is what makes it relevant. Pigeonhole cancellation itself requires no independence. Independence or a non-collapsed deformation is needed only to ensure that the resulting nonzero selector difference has a nonzero polynomial pair and analytic remainder. None of the statements proved here produces a sufficiently large primitive non-collapsed family, or proves that a collision has nonzero polynomial pair or nonzero remainder.

Example 3.7. At depths $R = S = 1$ the four-jet target is $(\mathbb{Z}/3\mathbb{Z})^2 \times (\mathbb{Z}/2\mathbb{Z})^2$, of cardinality $9 \cdot 4 = 36$, and the threshold reads $M \geq 4 \cdot 1 + 2 \cdot 1 = 6$. With six pairs there are $2^6 = 64$ binary selectors against 36 targets, so two of them collide and their difference is a vector in $\{-1, 0, 1\}^6$, not identically zero, killing all four jets.

Informally, the theorem says only this: once there are at least $4R + 2S$ pairs and the bottom depth R is positive, some coefficient vector in $\{-1, 0, 1\}^M$, not identically zero, kills all four jets of the corresponding combination. It does not say that the combination is nonzero as a pair of polynomials, and it does not say that its remainder is nonzero. Those are the two obligations Problem 8.1 carries.

One further exclusion is recorded here. It concerns the scalar parameters of Zudilin's cone rather than the coefficient polynomials.

Theorem 3.8 (scalar margin no-go). *Let $C_1 > 0$. If $C_0 \leq 0$ or $2C_0 \leq C_1$, then*

$$C_0 \log 3 - C_1 \log 2 < 0.$$

Written multiplicatively, the conclusion is $3^{C_0} < 2^{C_1}$. The inequality is immediate from $\log 3 < 2 \log 2$ and $C_1 > 0$, and is the checked [three-halves scalar margin](#). The interest is in the fence around it. The primary Zudilin theorem [5, Theorem 1, p. 154; Sec. 5, pp. 161–162] supplies an integer-base irrationality-exponent estimate on its parameter cone, and the elementary inequality $\mu \geq 2$ then forces $2C_0 \leq C_1$ whenever $C_0 > 0$; Lean checks that implication separately. The primary theorem assumes an integer base $p = 1/q$. It does not state a rational $p = 3/2$ theorem, so the all-scale coefficient construction and rational specialisation remain external to the checked result, which is the scalar parameter margin alone.

4 Failure of coordinatewise clearing at 3/2

This section and the next return to the elementary route and record what the residue s^n costs there. We first isolate the arithmetic that the clearing scheme leaves behind, in a form that does not mention the series. The point of doing so is that the clearing argument asks for two things at once, that a power of the numerator divide what has been accumulated and that what survives be small, and these are easier to play off against each other once the series has been discarded and only six natural numbers remain.

Definition 4.1 (coordinatewise corridor). Let a, b, N, K, Q, D be natural numbers. Say that (a, b, N, K, Q, D) is a *coordinatewise corridor* when

$$a > 0, \quad Q > 0, \quad D > 0, \quad D \leq N + K, \quad a^K \mid QD, \quad Qb^{N+K+1} < a^{K+1}.$$

The reading is: a and b are the numerator and denominator of the base, playing the roles of r and s in the introduction, so that $(a, b) = (3, 2)$ is the case of interest; N is the shift, K is the width of the cleared window, Q is the accumulated clearing factor, and D is the final coefficient being cleared. The bound $D \leq N + K$ is the only property of the coefficient used; for the divisor-counting coefficient it holds because $\tau(n) \leq n$. The divisibility $a^K \mid QD$ is the requirement that clearing succeeded coordinatewise, and the last inequality is the tail estimate that makes the trapped integer smaller than 1. The name records the shape of the constraint: the divisibility bounds a^K from above by $Q(N + K)$, the tail estimate bounds a^{K+1} from below by Qb^{N+K+1} , and admissible parameters must fit in the band between them. The definition is the [corridor predicate](#).

Theorem 4.2 (corridor bound). *If (a, b, N, K, Q, D) is a coordinatewise corridor, then*

$$b^{N+K+1} < a(N + K).$$

Proof. A routine divisibility computation. Since $QD > 0$ and $a^K \mid QD$, we have $a^K \leq QD$, and $D \leq N + K$ gives $a^K \leq Q(N + K)$. Hence

$$Qb^{N+K+1} < a^{K+1} = a^K \cdot a \leq Q(N + K) \cdot a,$$

and cancelling the positive factor Q gives the claim. $\square$

Formalised as the [power-versus-linear consequence](#).

The inequality of Theorem 4.2 is where the integer and rational cases part. At $b = 1$ it reads $1 < a(N + K)$, which holds for every $a \geq 2$ and every nonempty window; the corridor imposes no obstruction at all. At $b \geq 2$ the left side is exponential in $N + K$ and the right side is linear, so the corridor can survive only for small $N + K$. At $b = 2$ the crossing has already happened at the smallest admissible window.

Example 4.3. Take the smallest window, $N = K = 1$, and numerator $a = 3$. At $b = 1$ the tuple $(3, 1, 1, 1, 3, 1)$ is a corridor: $D = 1 \leq 2$, the divisibility reads $3 \mid 3$, and the tail estimate reads $3 \cdot 1^3 = 3 < 3^2 = 9$. At $b = 2$ no choice works. The tail estimate becomes $Q \cdot 2^3 < 3^2$, which forces $Q = 1$; the divisibility then reads $3 \mid D$, and the only candidates are $D = 1$ and $D = 2$, neither divisible by 3. This is the proof of Theorem 4.2 in miniature: it derives $a^K \leq Q(N + K)$, here $3 \leq 2$.

Proposition 4.4. *For every natural number $x \geq 2$ we have $3x < 2^{x+1}$.*

Proof. A routine induction from $x = 2$, where $6 < 8$. For the step, $2^{x+1} \geq 2^2 > 3$ when $x \geq 1$, so $3(x + 1) = 3x + 3 < 2^{x+1} + 2^{x+1} = 2^{x+2}$. $\square$

Formalised as the [exponential comparison](#).

Theorem 4.5 (no corridor at base 3/2). *For all $N \geq 1$ and $K \geq 1$ and all natural Q, D , the tuple $(3, 2, N, K, Q, D)$ is not a coordinatewise corridor.*

Proof. A corridor would give $2^{N+K+1} < 3(N + K)$ by Theorem 4.2, contradicting Proposition 4.4 applied to $x = N + K \geq 2$. $\square$

Formalised as the [corridor exclusion at three halves](#).

Theorem 4.5 excludes the coordinatewise clearing scheme at 3/2, and nothing else: it does not bound the denominator of $F(3/2)$, it does not show that $F(3/2)$ is irrational, and it does not show that $F(3/2)$ is rational. It also does not cover a clearing scheme of a different shape, since the corridor fixes one divisibility pattern and one tail inequality.

5 The cleared-tail recurrence and the size of the forcing term

Theorem 4.5 says that one scheme fails. This section identifies the quantity responsible, as an exact recurrence.

Let r, s, B, F be rationals with $r \neq 0$ and let $c : \mathbb{N} \rightarrow \mathbb{Q}$ be arbitrary. Define the prefix and the cleared tail state by

$$P_N = \sum_{m=0}^{N-1} \frac{c(m+1) s^{m+1}}{r^{m+1}}, \quad U_N = B r^N (F - P_N). \quad (*)$$

Thus P_N is the partial sum of $\sum_{n \geq 1} c(n)(s/r)^n$ through level N , and U_N is the tail of a putative value F after that level, scaled by $B r^N$. These are the [rational-base prefix](#) and the [cleared tail state](#).

Theorem 5.1 (cleared-tail recurrence). *Let $r, s, B, F \in \mathbb{Q}$ with $r \neq 0$, let $c : \mathbb{N} \rightarrow \mathbb{Q}$, and let P_N and U_N be as in $(*)$. Then for every N ,*

$$U_{N+1} = r U_N - B c(N+1) s^{N+1}.$$

Proof. An immediate computation. Expanding $P_{N+1} = P_N + c(N+1)s^{N+1}/r^{N+1}$ and $r^{N+1} = r^N \cdot r$ in the definition of U_{N+1} and clearing the denominator r^{N+1} , which is nonzero, gives the identity. $\square$

Formalised as the [cleared-tail recurrence](#).

The recurrence has a linear part $r U_N$ and a forcing term $B c(N+1) s^{N+1}$, the inhomogeneous term the state receives at each step. The integer-base argument works by keeping the state in a bounded window, and the forcing term is what it must absorb. Its size is what separates the two cases.

Theorem 5.2 (the forcing term). *Let s, B be natural numbers and $c : \mathbb{N} \rightarrow \mathbb{N}$, and put $G_N = B c(N+1) s^{N+1}$.*

1. *If $s \geq 2$, $B \geq 1$ and $c(N+1) \geq 1$, then $2^{N+1} \leq G_N$.*
2. *If $s = 1$, then $G_N = B c(N+1)$.*

Proof. Both parts are routine. For the first, $2^{N+1} \leq s^{N+1} = 1 \cdot s^{N+1} \leq B c(N+1) s^{N+1}$, using $B c(N+1) \geq 1$. The second part is the definition with $s = 1$. $\square$

Formalised as the [exponential lower bound](#) and the [integer-base collapse](#).

At $s = 1$ the forcing term is $B c(N+1)$, so it grows only as fast as the coefficient; for the divisor function that is $O(N^\epsilon)$ for every $\epsilon > 0$, and a bounded-state argument has room. At $s \geq 2$ the same term is at least 2^{N+1} whenever the coefficient is nonzero.

Example 5.3. Take $B = 1$, $c = \tau$ and $N = 9$, so that the coefficient is $\tau(10) = 4$. At $s = 1$ the forcing term is 4. At $s = 2$ it is $4 \cdot 2^{10} = 4096$, and part (1) of Theorem 5.2 already guarantees at least $2^{10} = 1024$ without knowing the coefficient at all.

This is an exact lower bound on one quantity, and it is all that is proved. It is not a proof that no bounded-state argument exists at $s \geq 2$; the theorem that one particular scheme fails is Theorem 4.5, and the bound here records the size of the term that scheme would have to absorb.

6 The height criterion at 7/2

In 1994 Bundschuh and Väänänen proved an irrationality criterion for a family of rational bases cut out by a height condition [4, Theorem 2, pp. 176–177]. We keep their notation: q is the base, and α and λ are the parameters of the criterion. In its special case $\alpha = -1$, the printed hypothesis is

$$\lambda < \left(\frac{1}{2} + \frac{1}{\pi^2} \right)^{-1}.$$

At $q = 7/2$ the Archimedean parameter is $\lambda = \log 7 / \log(7/2)$. The criterion therefore applies once the following strict inequality is checked.

Theorem 6.1 (the 7/2 height condition).

$$\frac{\log 7}{\log(7/2)} < \left(\frac{1}{2} + \frac{1}{\pi^2} \right)^{-1}.$$

Numerically the two sides are 1.5533 ... and 1.6630 ..., so the condition holds with a margin of about 0.11. The Lean proof factors the estimate into the [integer certificate](#) $2^{18} < 7^7$, the resulting [logarithmic ratio bound](#) $\log 2 / \log 7 < 7/18$, the [\$\pi\$ -bound](#) $1/\pi^2 < 1/9$, and the [strict margin](#)

$$\frac{\log 2}{\log 7} < \frac{1}{2} - \frac{1}{\pi^2}.$$

The [final height inequality](#) then rewrites $\log(7/2) = \log 7 - \log 2$ and closes the displayed condition.

This formalises the complete elementary parameter check at $q = 7/2$; it does not formalise Bundschuh and Väänänen’s analytic irrationality theorem, whose proof occupies pp. 189–193 of the source. The conclusion that $F(7/2)$ is irrational is consequently cited from that theorem, not claimed as a Lean theorem here.

The criterion does not cover 3/2, and the results of Sections 4 and 5 give no evidence either way about whether $F(3/2)$ is irrational.

7 Denominator exponents for a homogenised Padé construction

A second external route builds the linear forms of Section 2 by Padé approximation. Such a construction produces its coefficients as sums of rational summands, and to obtain integer rows one multiplies through by a single common denominator. The book-keeping obligation is then to check that no summand needs a larger denominator than

the one proposed, which is a comparison between the exponents alone. This section does that comparison, and only that.

No source construction is named here for the exponent expressions below, and they are checked as displayed polynomial identities rather than derived. Giving them Padé-theoretic force would additionally require deriving the expressions from a stated homogenised coefficient formula, proving integrality of the coefficients after multiplication by the proposed common denominator, and establishing nonvanishing and decay of the associated remainder. None of those three is claimed here.

For a homogenised construction over integer parameters the proposed common denominator exponent is $E_n = (3n^2 - n)/2$. Since only doubled exponents occur below, every statement lives over $\mathbb{Z}$ and no parity bookkeeping is needed; we write $\tilde{E}_n = 2E_n = 3n^2 - n$.

Proposition 7.1 (summand bound and exact gap). *Let $\tilde{E}_n = 3n^2 - n$ and put*

$$\tilde{P}(n, k) = 2(k(n - k) + nk) + k(k - 1),$$

$$\tilde{Q}(n, m) = 2(n^2 - n) + j^2 + 2jm + j - m^2 + 3m, \quad j = n - m - 1.$$

Then, for integers n, k, m :

1. if $0 \leq k \leq n$, then $\tilde{P}(n, k) \leq \tilde{E}_n$, and the gap factors as $\tilde{E}_n - \tilde{P}(n, k) = (n - k)(3n - k - 1)$;
2. $\tilde{E}_n - \tilde{Q}(n, m) = 2(n + m(m - 1))$ identically;
3. if $n \geq 0$ and $m \geq 1$, then $\tilde{Q}(n, m) \leq \tilde{E}_n$.

Part (1) is the [summand exponent bound](#), part (2) the [exact gap identity](#), and part (3) the [maximal exponent bound](#). The gap in (2) is an identity in $\mathbb{Z}[n, m]$, so (3) follows from $m(m - 1) \geq 0$ and $n \geq 0$; the hypothesis $m \geq 1$ names the range in which the corresponding summand does not vanish, and it is not the sharpest hypothesis under which the inequality holds.

Example 7.2. At $n = 2$ the proposed doubled exponent is $\tilde{E}_2 = 10$, and $\tilde{P}(2, k)$ takes the values 0, 6, 10 at $k = 0, 1, 2$. The three gaps are 10, 4, 0, matching the factorisation $(2 - k)(5 - k)$ of part (1); the summand at $k = 2$ is the one that saturates the proposed denominator. For part (2), at $m = 1$ we have $j = 0$ and $\tilde{Q}(2, 1) = 6$, with gap $4 = 2(2 + 1 \cdot 0)$.

These are routine inequalities between polynomials in the exponents. They establish that the proposed exponent $\tilde{E}_n$ dominates the two displayed summand exponent expressions, and nothing further. Positivity of the remainder, its rate of decay, and the comparison of that rate against the denominator height are the analytic obligations, and none of them is treated here, so nothing in this section is an irrationality measure.

8 Complements and further questions

Problem #1049 is open. The following is the exact statement that a construction at 3/2 would have to supply. Theorem 3.6 supplies the combinatorial step: at depths R, S with

$R > 0$, once at least $4R + 2S$ pairs are in hand, a coefficient vector on which all four jets vanish exists. What the problem adds is that those pairs be primitive-normalised Heine–Zudilin pairs drawn from a family that does not degenerate, that the depths grow quadratically, and that the resulting remainder be nonzero with a local gain beating the Archimedean height.

Problem 8.1 (growing-rank simultaneous endpoint-jet kernel). For positive quadratic target depths R_n, S_n , exhibit at least $4R_n + 2S_n$ *primitive-normalised* Heine–Zudilin coefficient pairs from a genuinely non-collapsed deformation, and prove that at least one resulting checked collision satisfies

$$J_{3,R_n}\left(\sum_j \lambda_j U_j\right) = J_{3,R_n}\left(\sum_j \lambda_j V_j\right) = 0, \quad J_{2,S_n}\left(\sum_j \lambda_j U_j\right) = J_{2,S_n}\left(\sum_j \lambda_j V_j\right) = 0,$$

where R_n, S_n have the required quadratic order, while the resulting remainder is nonzero and its local gain beats the Archimedean height. The checked collision already has coefficients in $\{-1, 0, 1\}$. A rank-saturated contiguous-shift family or a genuinely independent deformation would be a first candidate.

Rescaling a row along a ray, letting the row contents grow, or computing a further common divisor of the two specialised evaluations does not address the obstruction that survives Corollary 3.5; each is one of the two devices excluded there, and what Problem 8.1 asks for is additive. Devices outside that corollary are not addressed either way: inserting a multiplicative cyclotomic factor changes the coefficient polynomials rather than scaling a row, and cross-row common factors, determinant-specific arithmetic, additive combinations, and a different architecture all lie outside the two devices it excludes. A theorem showing that every available family is too small or too degenerate to meet the requirements of Problem 8.1 would show that no argument of this shape can succeed.

The unrestricted question, which rational bases give an irrational value, remains open, and is not reduced to the problem above.

Statements and declarations

Artefact and data availability. The [pinned formal-source revision](#) contains the Lean sources, the fixed toolchain, and the library manifest used in the verification. This manuscript provides navigation rather than proof authority.

Declaration of generative AI use. Large-language-model agents were used throughout development to draft and revise prose, formal proofs, and software. The author set the objectives and acceptance criteria, selected and reviewed the claims, and approved the published version. The author assumes responsibility for the accuracy, interpretation, and presentation of the work. Generative systems are production tools; they are not authors and supply no independent authority. Lean checks each proof term against the fixed library version, and the sources linked here contain no proof placeholders and no project-defined axioms; Lean does not authorise the exposition, the citation choices, or the interpretation, for which the author remains responsible.

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A Guide to the formal sources

Each linked phrase opens its Lean declaration at the pinned source revision 08d83b6689c8. The declarations of this note live in three modules: the corridor, the cleared-tail recurrence, and the two arithmetic certificates in the first; the Padé exponent arithmetic in the second; and the Heine–Zudilin endpoint arithmetic in the third. The link coordinates are validated against that pinned revision, so they remain correct as later work moves lines in the working tree.

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